Materials Science 1- Fundamentals of crystallography

Learning objectives and Learning outcomes

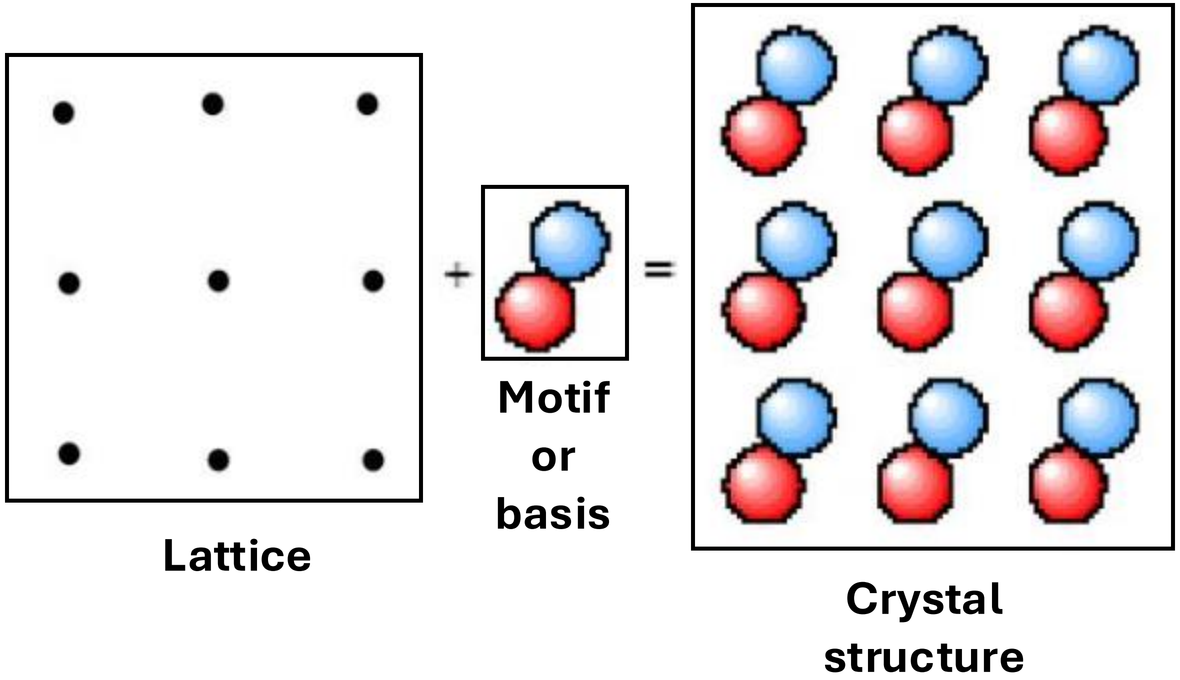
- Identify crystal systems and Bravais lattices and unit cells in 3D;
- Identify close-packed structures;
- Identify point coordinates, crystallographic directions and crystallographic planes.

Fundamentals of crystallography

Crystal structure: can be constructed by placing atoms or groups of atoms or molecules (motif) on each lattice point.

Crystal structure in 2D:

$$\boxed{Crystal = lattice + motif}$$

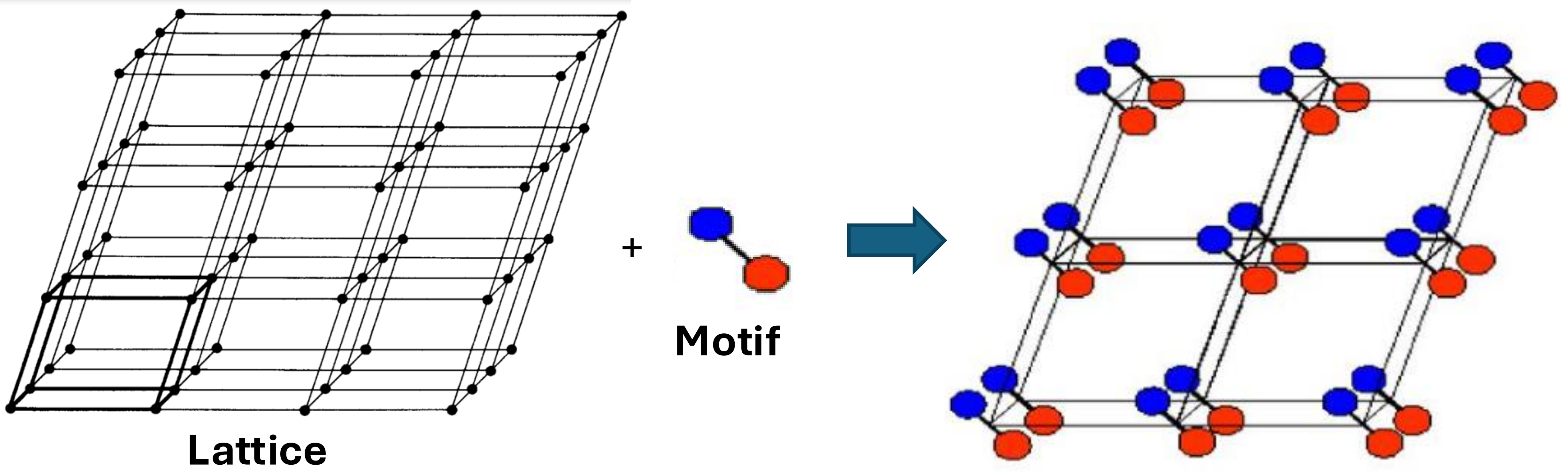


Motif or basis of crystal: atoms or groups of atoms or molecules associated with each lattice point.

Fundamentals of crystallography

<u>Crystal structure in 3D:</u>

$$\boxed{Crystal = lattice + motif}$$



Fundamentals of crystallography

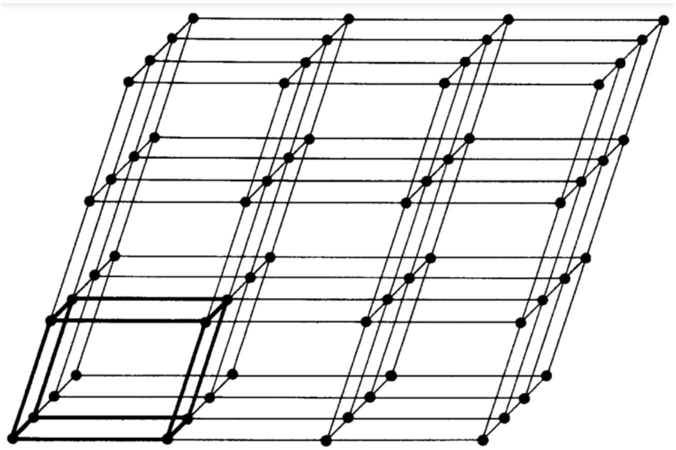
<u>Crystal structure:</u> can be constructed by placing atoms or groups of atoms or molecules (having a particular motif) on each lattice point.

$$\boxed{Crystal = lattice + motif}$$

<u>Lattice:</u> underlying symmetry of the crystal (How to repeat)

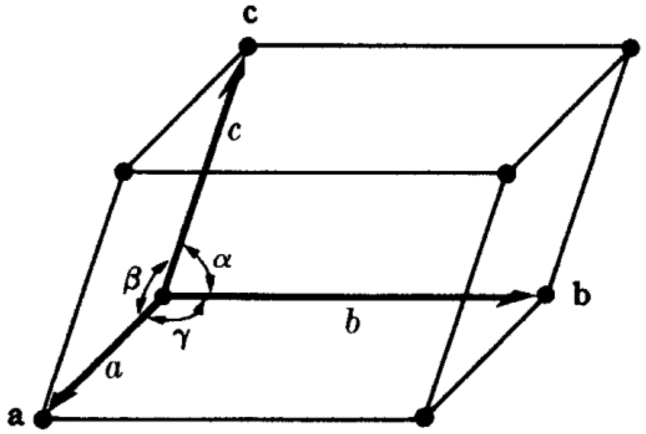
<u>Motif:</u> atoms or group of atoms which are periodically repeated (What to repeat)

Fundamentals of crystallography



3D Point lattice

The cells of the point lattice are identical, and can be considered as "**unit cells**".



Unit cell

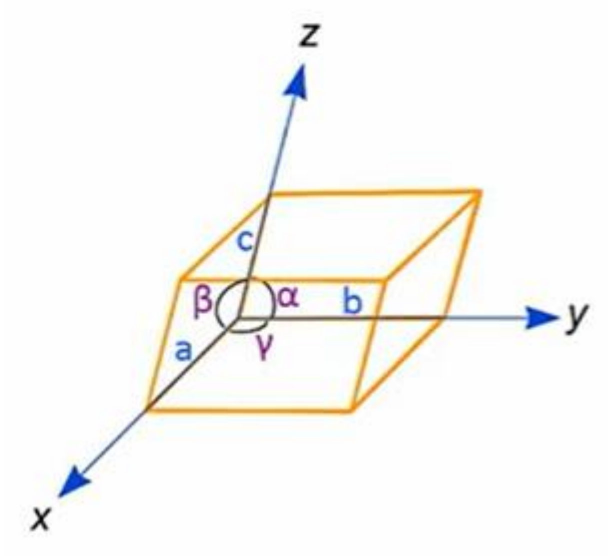
<u>Unit cells in 3D lattices:</u> unit elements representative of the entire lattice. Described by the three vectors a, b, and c drawn from one corner of the cell taken as origin.

These vectors define the cell and are called the **crystallographic axes** of the cell. They can be described in terms of their lengths (a, b, c) and the angles between them (α, β, y) that represent the complete set of **lattice constants or lattice parameters** of the unit cell.

Fundamentals of crystallography

Features of unit cells:

Unit cells are characterized by three main features: metrics, symmetry and chemical compositions.



Metric

- it is defined by six cell/lattice parameters
  - the three cell/lattice constants, the lengths of the edges (a, b, and c)
  - and three angles between these edges (α, β, and γ)

Symmetry

- it contains all present symmetry elements
- defines the minimum size of the unit cell

Chemical Composition (Stoichiometry)

- the chemical content of an unit cell corresponds to the chemical composition of the considered compound (!)

Fundamentals of crystallography

Crystal systems

By changing the lattice parameters (length and angle), it is possible to produce unit cells of various shapes. It turns out that only seven different types of geometrical unit cells are needed to map all the possible point lattices. These correspond to the **7 crystal systems**.

Symmetry

restrictions for	*cell constants*	*cell angles*
triclinic	none	none
monoclinic	none	$\alpha = \gamma = 90°$
orthorhombic	none	$\alpha = \beta = \gamma = 90°$
tetragonal	a = b	$\alpha = \beta = \gamma = 90°$
trigonal	a = b	$\alpha = \beta = 90°; \gamma = 120°$
hexagonal	a = b	$\alpha = \beta = 90°; \gamma = 120°$
cubic	a = b = c	$\alpha = \beta = \gamma = 90°$

Trigonal and hexagonal have the same restrictions since they belong to the same crystal family. However, crystal systems are classified based on symmetry, and trigonal and hexagonal systems can have different degree of symmetry.

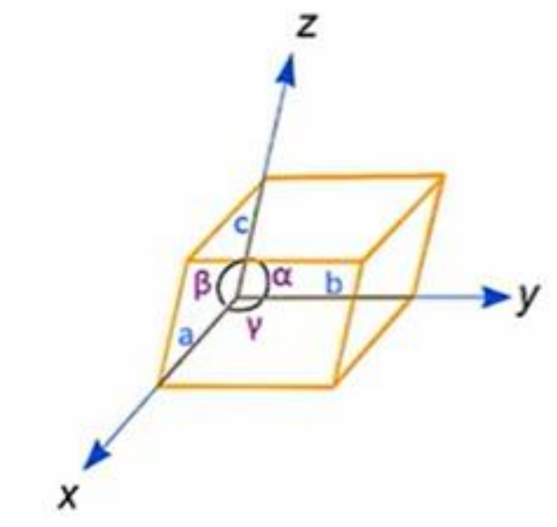
The 7 crystal systems.

Crystal system	Conventional coordinate system	
	Restrictions on cell parameters	Parameters to be determined
Triclinic	None	$a, b, c,$ α, β, γ
Monoclinic	*b*-unique setting $\alpha = \gamma = 90°$	a, b, c β‡
	c-unique setting $\alpha = \beta = 90°$	$a, b, c,$ γ‡
Orthorhombic	$\alpha = \beta = \gamma = 90°$	a, b, c
Tetragonal	$a = b$ $\alpha = \beta = \gamma = 90°$	a, c
Trigonal	$a = b$ $\alpha = \beta = 90°, \gamma = 120°$	a, c
	$a = b = c$ $\alpha = \beta = \gamma$ (rhombohedral axes, primitive cell) $a = b$ $\alpha = \beta = 90°, \gamma = 120°$ (hexagonal axes, triple obverse cell)	a, α
Hexagonal	$a = b$ $\alpha = \beta = 90°, \gamma = 120°$	a, c
Cubic	$a = b = c$ $\alpha = \beta = \gamma = 90°$	a

Fundamentals of crystallography

Crystal family	Crystal system	Required symmetries of the point group	Point groups	Space groups
Triclinic	Triclinic	None	2	2
Monoclinic	Monoclinic	1 twofold axis of rotation or 1 mirror plane	3	13
Orthorhombic	Orthorhombic	3 twofold axes of rotation or 1 twofold axis of rotation and 2 mirror planes	3	59
Tetragonal	Tetragonal	1 fourfold axis of rotation	7	68
Hexagonal	Trigonal	1 threefold axis of rotation	5	7
				18
	Hexagonal	1 sixfold axis of rotation	7	27
Cubic	Cubic	4 threefold axes of rotation	5	36
6	7	**Total**	32	230

Fundamentals of crystallography



Warning!

When classifying crystal systems, several textbooks and notes use the following table, with unnecessary restrictions on the lattice parameters.

	cell constants	*angles*
triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$
monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90°, \beta \neq 90°$
orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$
tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$
trigonal	$a = b \neq c$	$\alpha = \beta = 90°; \gamma = 120°$
hexagonal	$a = b \neq c$	$\alpha = \beta = 90°; \gamma = 120°$
cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$

According to expert crystallographers, this table is incorrect.

- M. Nespolo: “A number of recent (and not so recent) textbooks introduce the concept of crystal system on the basis of the metric relations of the conventional unit cell, which is incorrect”. J. Appl. Cryst. (2015). 48, 1290–1298.
- G.Burns, M.Glazer: Remember that it is the symmetry that puts restrictions on the axes and interaxial angles, and not the other way around, and it is the least symmetrical property that imposes its symmetry as "crystal symmetry."

Space Groups for Solid State Scientists (page 23), second edition

Fundamentals of crystallography

VIMs (Very Important Messages)

- The cell parameters give only an indication of the underlying symmetry!
- It is not the metric which determines the symmetry!
- It is the other way round: the symmetry determines the metric, although not in a biunique way....
- Crystals, which belong to the triclinic crystal system (according to their symmetry), may have *coincidentally* cell parameters like this:

$$a = b = c \qquad \alpha = \beta = \gamma = 90°$$

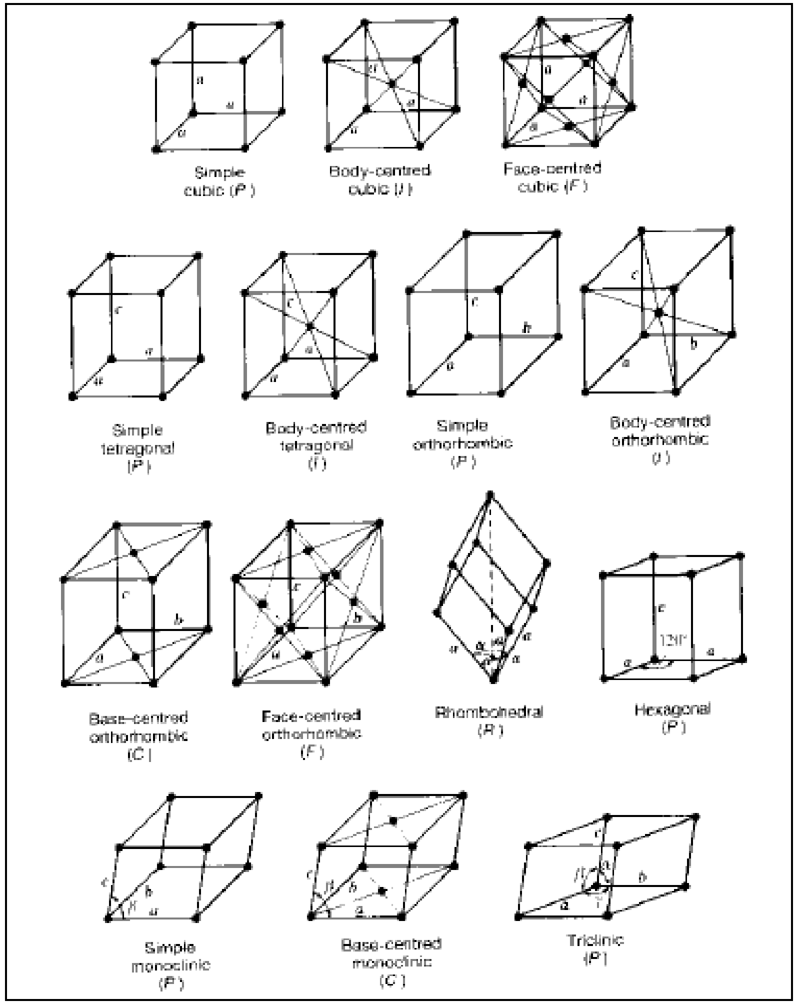
- Vice versa: A crystal of the cubic crystal system has (*necessarily* and due to the symmetry!) always the following cell parameters:

$$a = b = c \qquad \alpha = \beta = \gamma = 90°$$

Fundamentals of crystallography

3D Bravais lattices

Group theory has demonstrated that based on the symmetry elements of the unit cell and the definition of point lattice in 3D there are only 14 possible Bravais lattices (some of the 28 conceivable are redundant and some are incompatible for symmetry reasons):



Lattice system	Axial lengths and angles	Bravais lattice	Lattice symbol
Cubic	Three equal axes at right angles $a = b = c, \quad \alpha = \beta = \gamma = 90^{\circ}$	Simple Body-centered Face-centered	P I F
Tetragonal	Three axes at right angles, two equal $a = b \neq c, \quad \alpha = \beta = \gamma = 90^{\circ}$	Simple Body-centered	P I
Orthorhombic	Three unequal axes at right angles $a \neq b \neq c, \quad \alpha = \beta = \gamma = 90^{\circ}$	Simple Body-centered Base-centered Face-centered	P I C F
Rhombohedral	Three equal axes, equally inclined $a = b = c, \quad \alpha = \beta = \gamma \neq 90^{\circ}$	Simple	R
Hexagonal	Two equal coplanar axes at 120°, third axis at right angles $a = b \neq c, \quad \alpha = \beta = 90^{\circ}, \quad \gamma = 120^{\circ}$	Simple	P
Monoclinic	Three unequal axes, one pair not at right angles $a \neq b \neq c, \quad \alpha = \gamma = 90^{\circ} \neq \beta$	Simple Base-centered	P C
Triclinic	Three unequal axes, unequally inclined and none at right angles $a \neq b \neq c, \quad \alpha \neq \beta \neq \gamma \neq 90^{\circ}$	Simple	P

The symbol ≠ means that equality is not required by symmetry. Accidental equality may occur

Fundamentals of crystallography

Crystal family	Crystal system	Required symmetries of the point group	Point groups	Space groups	Bravais lattices	Lattice system
Triclinic	Triclinic	None	2	2	1	Triclinic
Monoclinic	Monoclinic	1 twofold axis of rotation or 1 mirror plane	3	13	2	Monoclinic
Orthorhombic	Orthorhombic	3 twofold axes of rotation or 1 twofold axis of rotation and 2 mirror planes	3	59	4	Orthorhombic
Tetragonal	Tetragonal	1 fourfold axis of rotation	7	68	2	Tetragonal
Hexagonal	Trigonal	1 threefold axis of rotation	5	7	1	Rhombohedral
				18	1	Hexagonal
	Hexagonal	1 sixfold axis of rotation	7	27		
Cubic	Cubic	4 threefold axes of rotation	5	36	3	Cubic
6	7	**Total**	32	230	14	7

Note: there is no "trigonal" lattice system. To avoid confusion of terminology, the term "trigonal lattice" is not used.

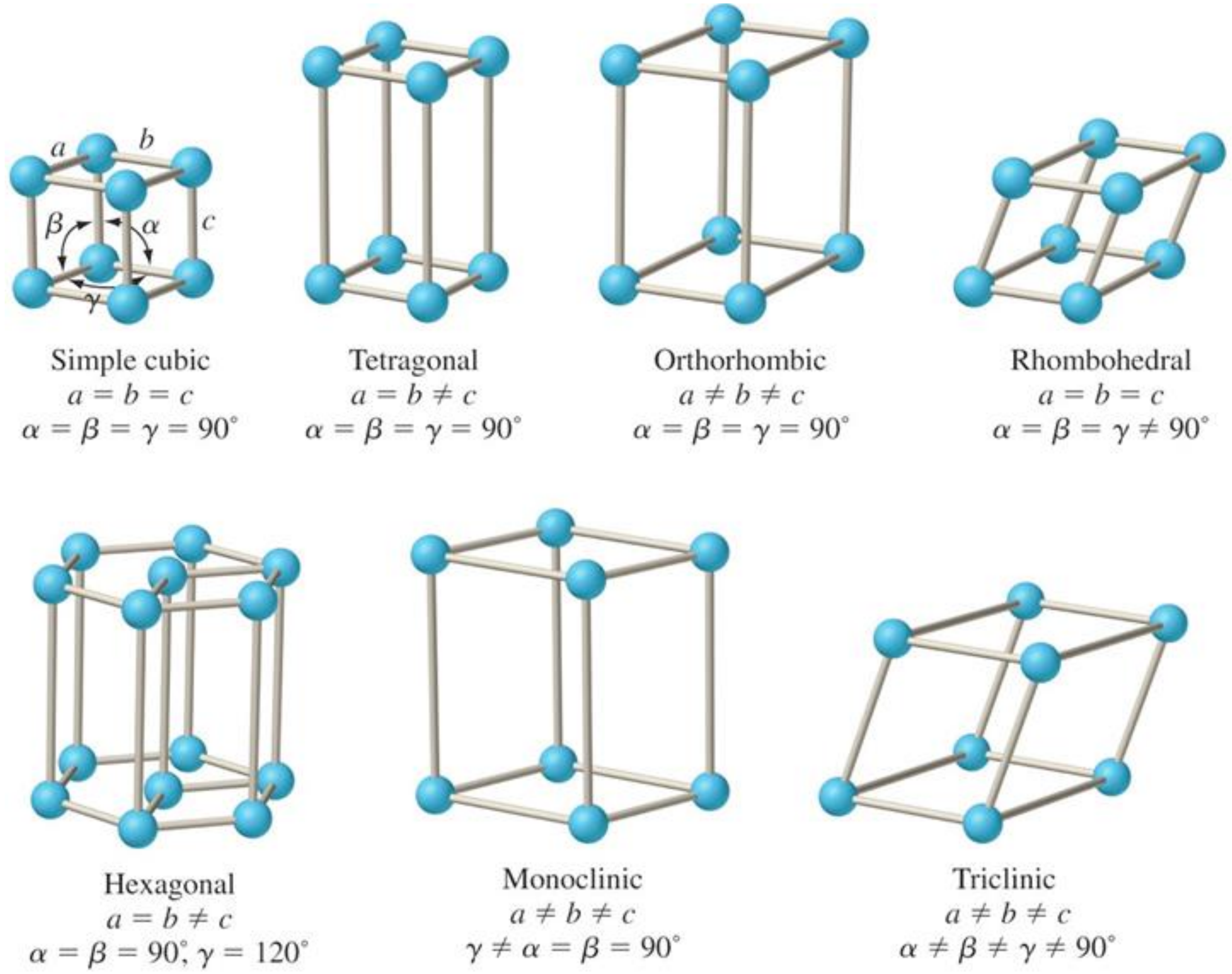
Fundamentals of crystallography

Unit cells of the 3D Bravais lattices

Unit cell shape	Unit Cell Type	Lattice Parameters	Angles
Cubic	Primitive, Body-centered, Face-centered	a = b = c	$\alpha = \beta = \gamma = 90°$
Tetragonal	Primitive, Body-centered	a = b	$\alpha = \beta = \gamma = 90°$
Orthorhombic	Primitive, Body-centered, Face-centered, Base-centered	None	$\alpha = \beta = \gamma = 90°$
Rhombohedral	Primitive	a = b = c	$\alpha = \beta = \gamma$
Hexagonal	Primitive	a = b	$\alpha = \beta = 90°$, $\gamma = 120°$
Monoclinic	Primitive, Base-centered	None	$\alpha = \gamma = 90°$
Triclinic	Primitive	None	None

Fundamentals of crystallography

<u>Geometry of unit cells of the 3D Bravais lattices</u>



Further readings

Fundamentals of crystallography

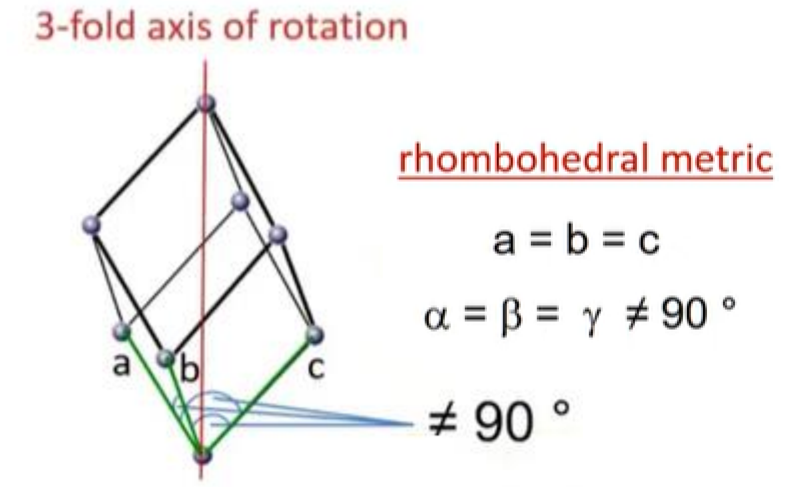
Relationship between trigonal, rhombohedral and hexagonal systems

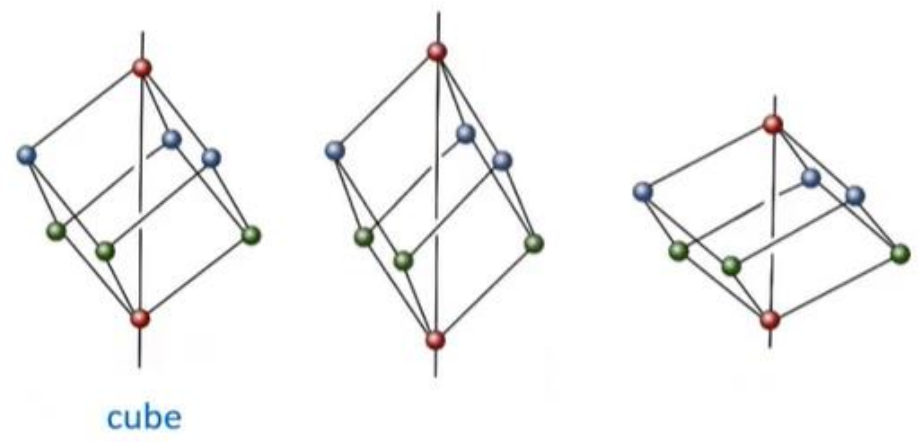
Trigonal crystal system

There are two types of crystals belonging to the trigonal system; however, both are characterized by 3-fold symmetry:

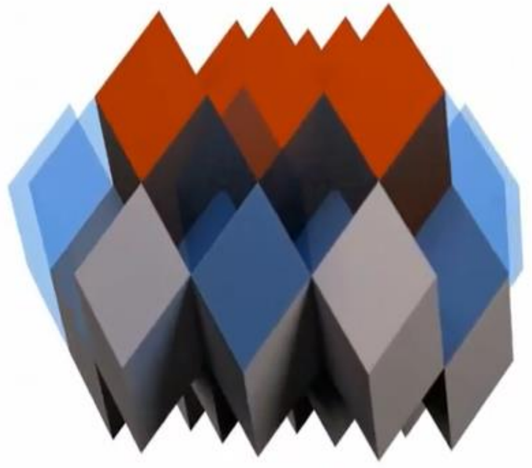
1. **Crystals with a rhombohedral primitive unit cell;**
2. **Crystals with a hexagonal primitive unit cell.**

Rhombohedral primitive unit cell





The rhombohedral unit cell can be obtained by a stretching or compression of a cube along the principle diagonal.

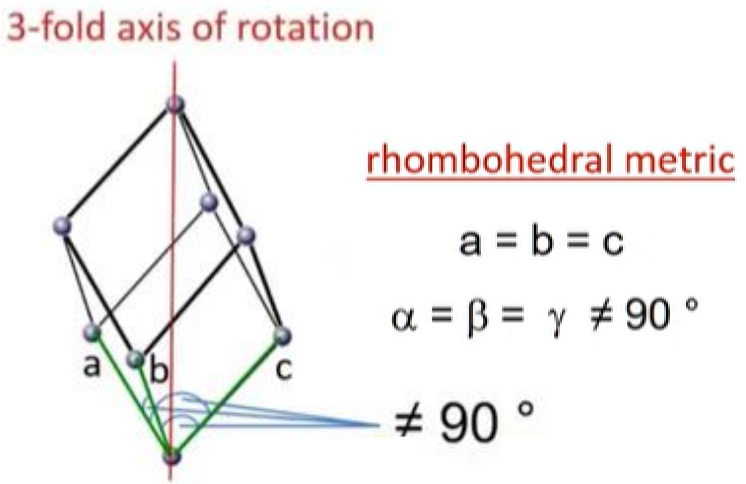


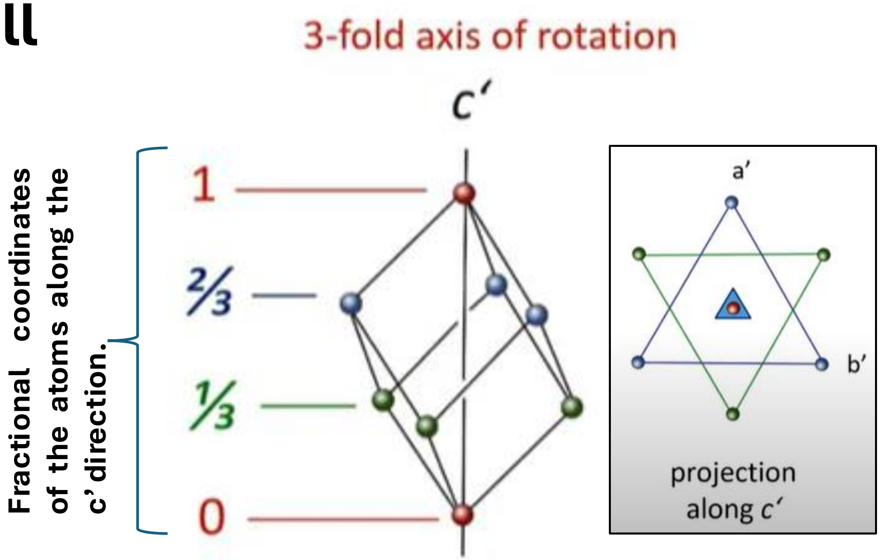
The rhombohedrons are parallelepipeds and so are completely space-filling.

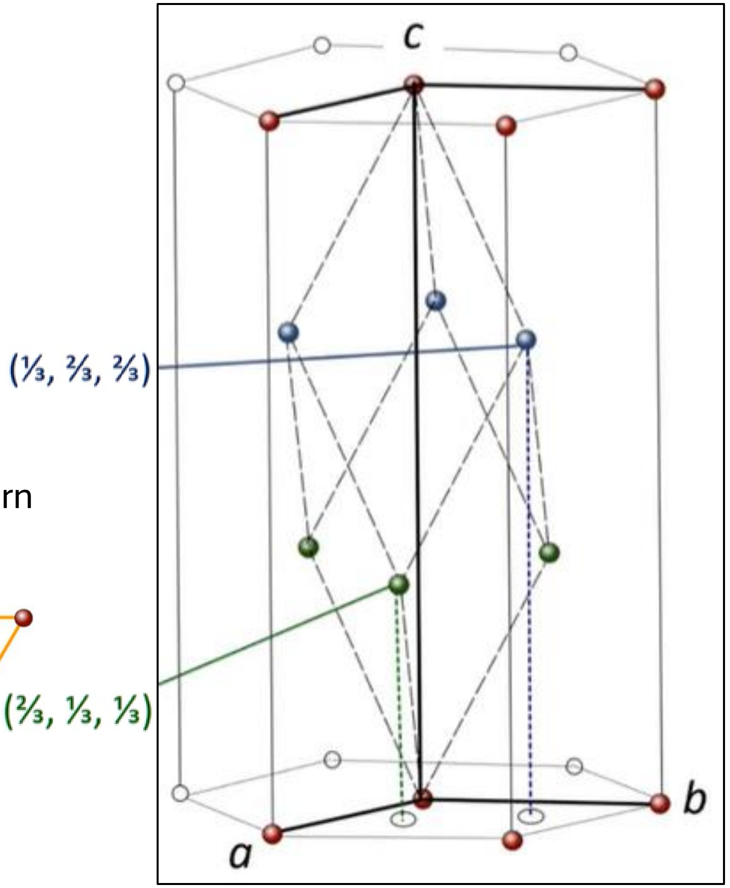
Fundamentals of crystallography

Relationship between trigonal, rhombohedral and hexagonal systems

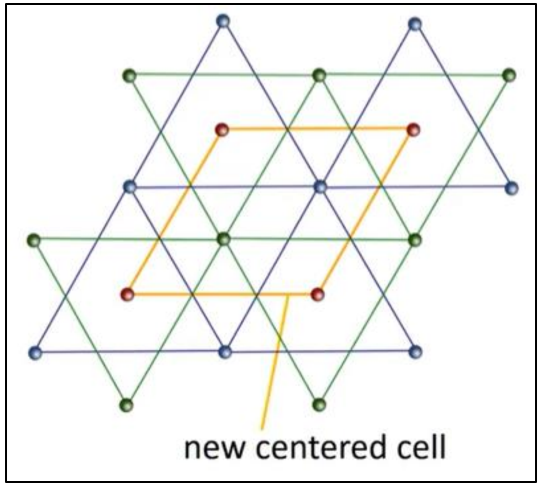
Rhombohedral primitive unit cell

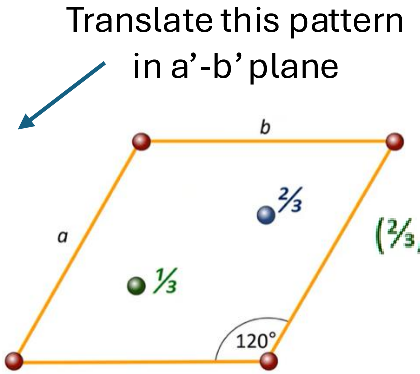






Notice that the 3-fold axis of rotation does not coincide with the direction of the lattice parameter c. In crystallography, the rotational axis of the highest order must be parallel to the c-axis. To fulfil this rule, another point lattice should be constructed that has the third direction coinciding with 3-fold axis of rotation.





The new unit cell has 2 additional lattice points and has a hexagonal metric: a=b; γ=120°.

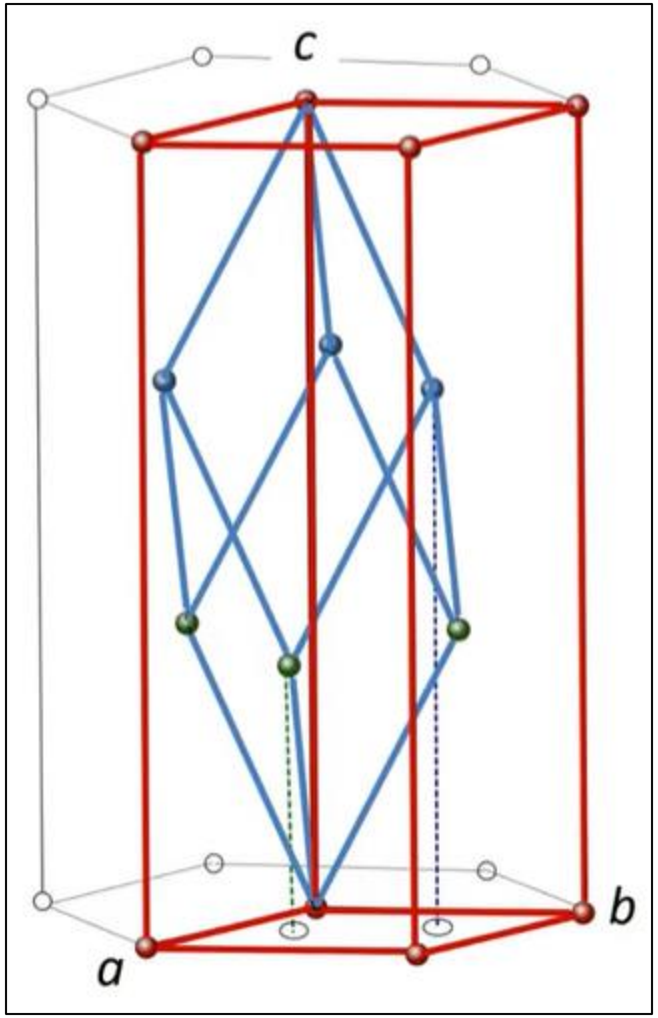
Relationship between the primitive rhombohedral cell and the hexagonal centred cell in which it is contained.

Fundamentals of crystallography

Relationship between trigonal, rhombohedral and hexagonal systems

Trigonal crystal system

- All crystals of the trigonal crystal system can be represented in a lattice with a hexagonal metric.
- This is also valid for trigonal crystals that have a rhombohedral lattice when they are represented in the primitive form (blue unit cell in the figure).
- It is always possible to transform the primitive rhombohedral cell into a hexagonal centred cell (red unit cell in the figure).
- Therefore, there are rhombohedral lattices but not rhombohedral crystal systems.



Fundamentals of crystallography

Relationship between trigonal, rhombohedral and hexagonal systems

In other words:

Why no trigonal lattices?

- Because the trigonal crystal system is defined by symmetry (threefold rotational symmetry), and the crystallographic convention is to use either rhombohedral or hexagonal lattices to describe trigonal symmetry. These lattice types can adequately describe all trigonal crystals, so a separate trigonal lattice isn't necessary.

- Rhombohedral lattice: Handles trigonal symmetry directly with a primitive unit cell.
- Hexagonal lattice: Describes trigonal symmetry using a non-primitive unit cell but is widely used due to its compatibility with sixfold and threefold symmetries.

- Thus, the absence of a "trigonal lattice" reflects how trigonal symmetry can be fully accommodated by rhombohedral or hexagonal lattices.

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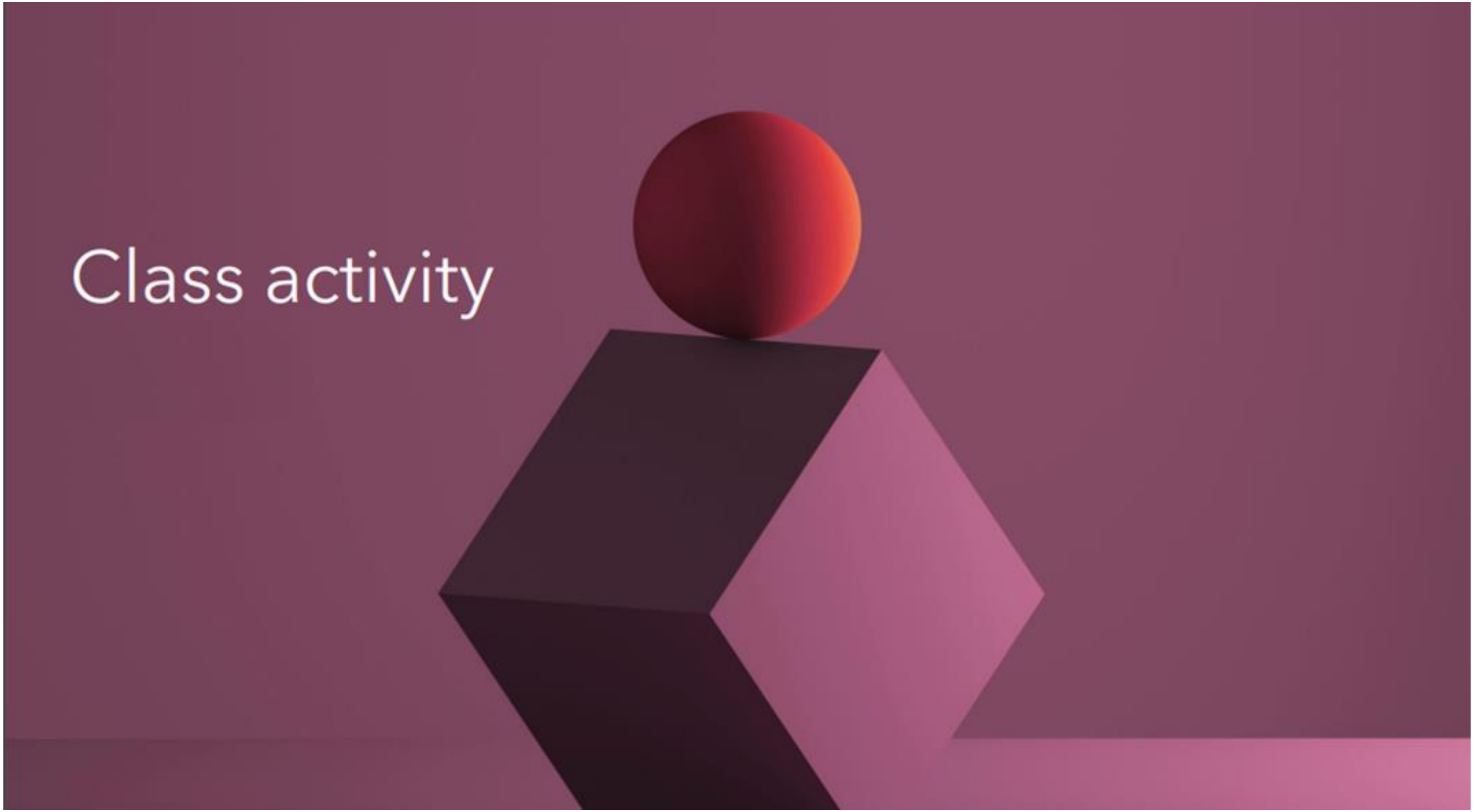
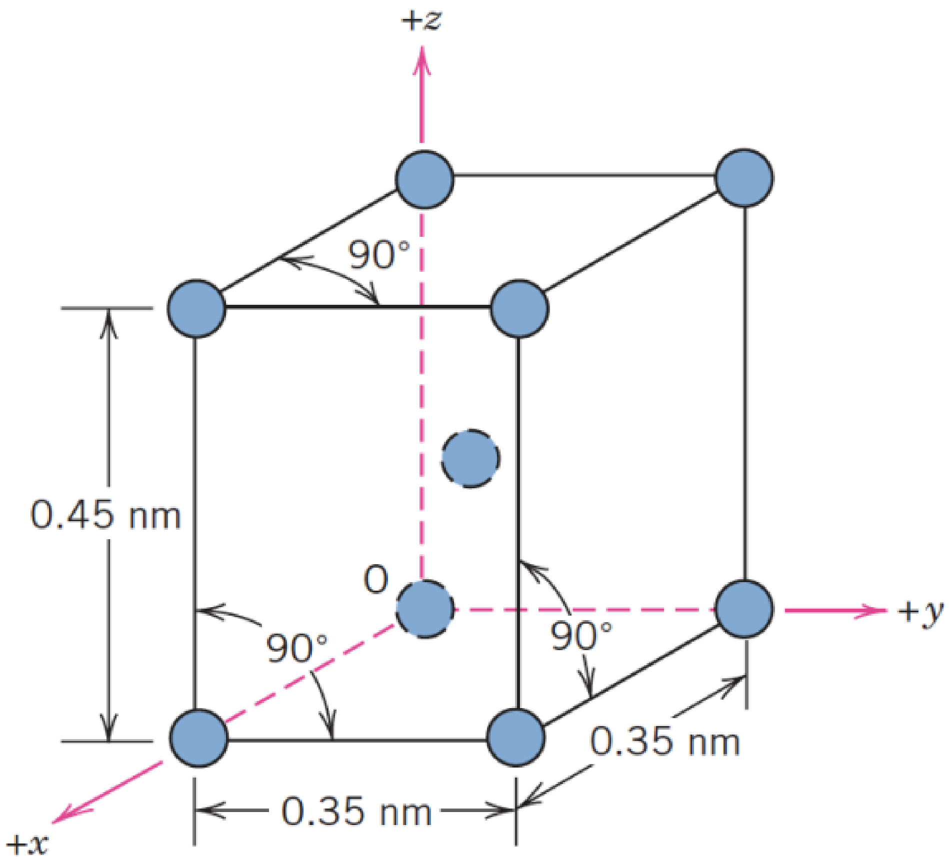


Fundamentals of crystallography

Crystal systems

Example 1: The figure below shows a unit cell for a hypothetical metal.



Answer the following questions:

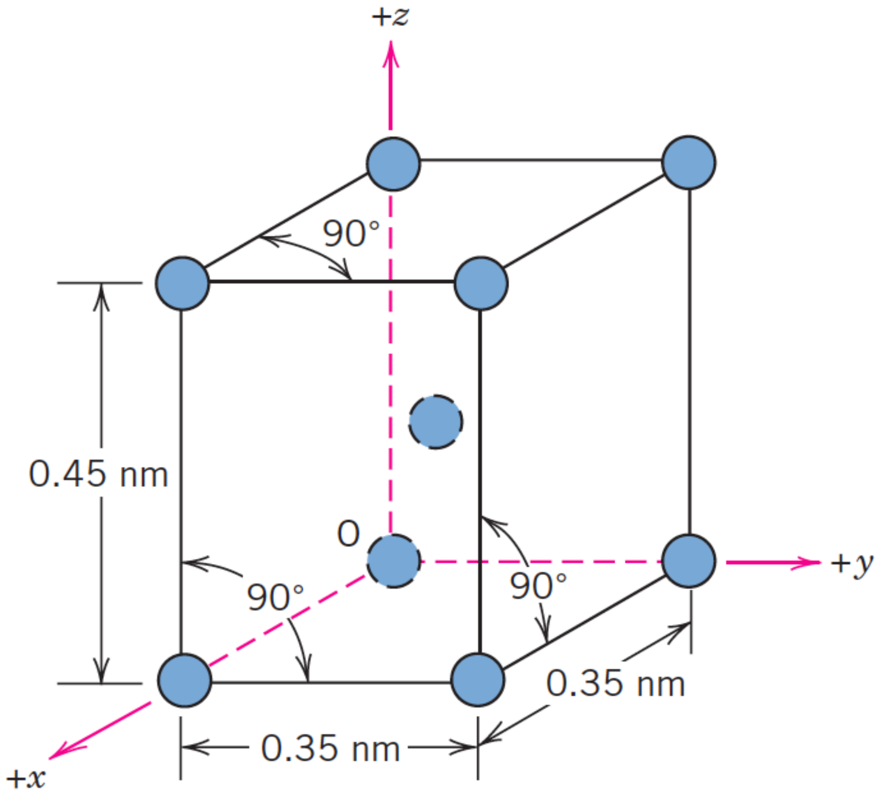
1) To which crystal system does this unit cell belong?

2) What would this crystal structure be called?

Fundamentals of crystallography

Crystal systems

Answer:



a. The structure is a tetragonal crystal system since *a* = *b* = 0.35 nm, *c* = 0.45 nm, and *a* = *b* = *g* = 90°.

b. The crystal structure would be called *body-centered tetragonal*.

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<u>Example 2:</u> Crystal systems.

- Sketch a unit cell for the face-centered orthorhombic crystal structure.
- Draw all the atoms in the cell unit.

Fundamentals of crystallography

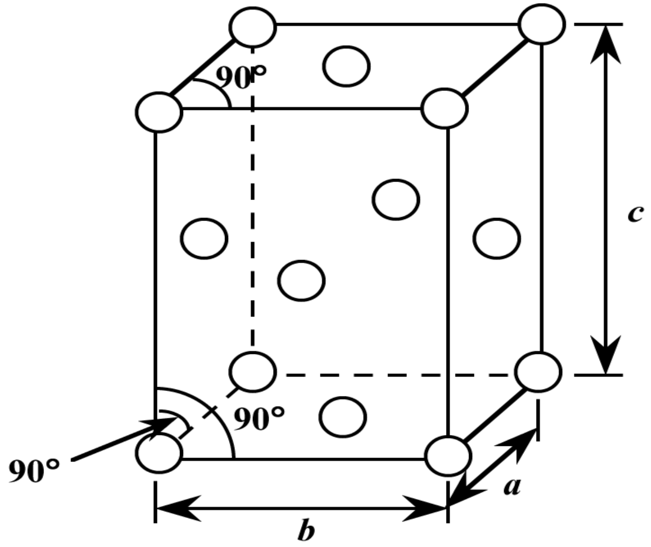
Example 2:

- Sketch a unit cell for the face-centered orthorhombic crystal structure.
- Draw all the atoms in the cell unit.

Answers:

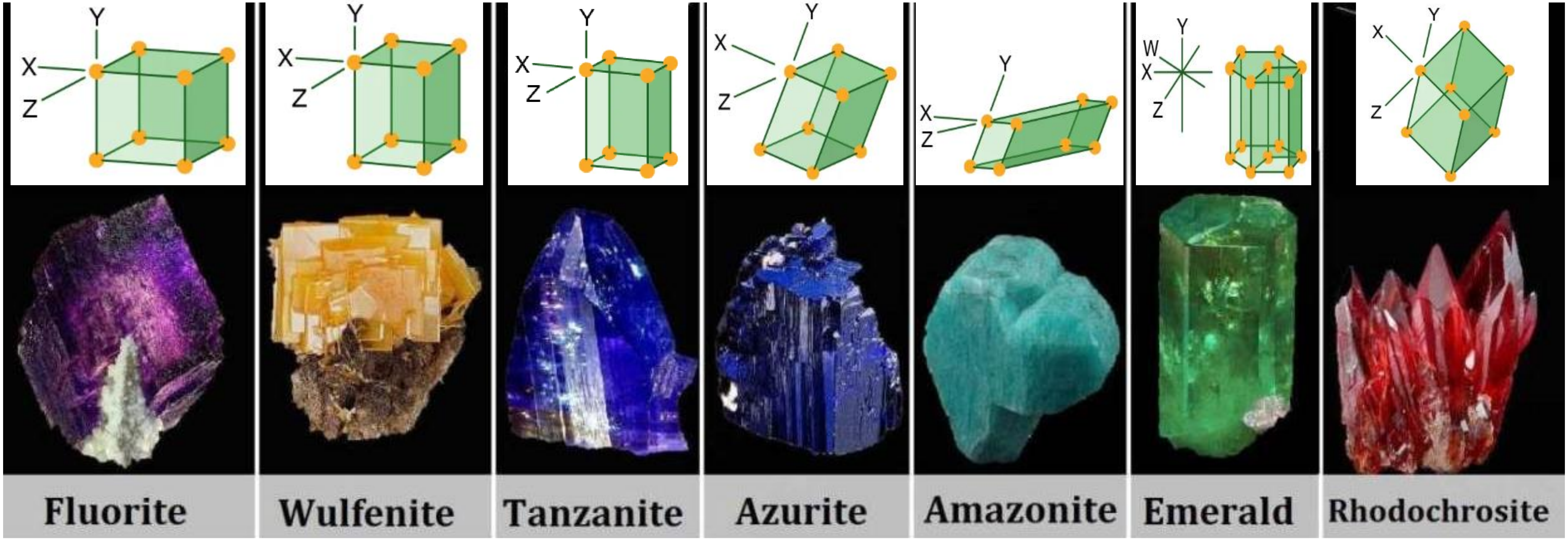
For the orthorhombic crystal system, no restriction on a, b, c so that $a \neq b \neq c$, and $\alpha=\beta=\gamma=90°$. Also, for the face-centered orthorhombic crystal structure, atoms will be located at the centers of all 6 faces (in addition to all 8 corners).

A drawing of the FC orthorhombic.



Fundamentals of crystallography

<u>Example 3:</u> Identify the crystal system for each one of the minerals in the picture



Fundamentals of crystallography

<u>Example 3:</u> Identify the crystal system for each one of the minerals in the picture

<u>Answer:</u>

Cubic **Tetragonal** **Orthorhombic** **Monoclinic** **Triclinic** **Hexagonal** **Trigonal**